

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

ITUE203: WEB DEVELOPMENT FRAMEWORKS

QUESTION BANK

Q1. Online Scholarship Portal

A university scholarship portal provides a search feature through which students can filter scholarships by programme and family-income range. The same portal also allows students to submit an application containing personal details, bank-account information, and uploaded evidence.

- a) Select GET or POST for the scholarship-search operation and for the application-submission operation (2 marks)
- b) Justify both choices with reference to visibility of data, bookmarking, request purpose, and server-side processing (3 marks)

Q2. Hospital Appointment System

A hospital website uses the following URL after a patient submits an appointment request:

<https://carehospital.org/book?patient=Riya&mobile=9876543210&symptom=chest-pain>

- a) Identify two problems with transmitting the appointment information in this manner (2 marks)
- b) Recommend a more suitable HTTP method and explain why it is appropriate for this operation (3 marks)

Q3. Online Examination Portal

An online examination portal has two operations: (i) viewing a publicly available examination timetable, and (ii) submitting a student's final answers when the examination ends.

- a) Choose the appropriate HTTP method for each operation (2 marks)
- b) Explain the difference between the selected methods in terms of data location, repeatability, URL visibility, and effect on server data (3 marks)

Q4. Employee Performance Report

A company wants to display quarterly employee performance in which Technical Score and Communication Score are shown under both Mid-Year Review and Final Review. Study the required structure below and write the complete HTML table code.

EMPLOYEE PERFORMANCE REPORT						
Employee ID	Name	Mid-Year Review		Final Review		Status
		Technical	Communication	Technical	Communication	
E01	Aarav	78	82	84	86	Confirmed
E02	Meera	81	79	88	83	Confirmed

- a) Write the complete HTML code using table, tr, th, and td elements (3 marks)
- b) Use appropriate rowspan and colspan values to reproduce the grouped headings exactly (2 marks)

Q5. University Laboratory Schedule

The Department of Information Technology wants to publish a laboratory schedule. The Batch column must span two practical sessions for each batch, and the course title must span all columns.

IT LABORATORY SCHEDULE				
Batch	Day	Time	Laboratory	Faculty
B1	Monday	10:00-11:00	Lab-1	Prof. Shah
	Thursday	02:00-03:00	Lab-2	Prof. Patel
B2	Tuesday	11:00-12:00	Lab-1	Prof. Mehta
	Friday	03:00-04:00	Lab-2	Prof. Rao

a) Write the complete HTML code for the displayed schedule **(3 marks)**

b) Use colspan for the title and rowspan for the B1 and B2 batch cells **(2 marks)**

Q6. Restaurant Order Summary

A restaurant wants to display item-wise orders grouped under Food and Beverage categories. The category name should appear only once and vertically span all corresponding items.

CUSTOMER ORDER SUMMARY				
Category	Item	Quantity	Unit Price	Amount
Food	Paneer Wrap	2	₹180	₹360
	Veg Pizza	1	₹300	₹300
	French Fries	2	₹120	₹240
Beverage	Cold Coffee	2	₹150	₹300
	Fresh Lime	1	₹90	₹90

a) Write the complete HTML table code **(3 marks)**

b) Use one colspan and the required rowspan values to reproduce the structure **(2 marks)**

Q7. Downloading an E-book from an Online Library

A student searches for an e-book in the university library portal and clicks the Download PDF button. After authentication, the PDF is delivered to the browser.

Explain the complete HTTP request-response cycle for this operation. Your answer should cover:

- request initiation
- DNS resolution
- connection with the server
- authentication and server-side file retrieval
- HTTP response headers and PDF body
- browser download or display

Q8. Viewing a Personalized Examination Result

A student logs in to the result portal and selects Semester 3. The server retrieves the student's marks from a database and displays a personalized result page.

Explain the complete HTTP request-response cycle for this operation. Your answer should cover:

- browser request
- DNS and secure connection
- session or credential verification
- database processing
- HTTP status and response content
- browser rendering

Q9. Booking a Cinema Ticket

A customer selects a movie, chooses seats, and clicks Confirm Booking. The server checks seat availability, records the booking, and returns a confirmation page.

Explain the complete HTTP request-response cycle for this operation. Your answer should cover:

- form submission
- request method and request body
- server validation
- database update
- success or failure response
- confirmation-page rendering

Q10. Department Event Page

Consider the following HTML code:

```
<!DOCTYPE html>
<html>
<head>
  <title>Department Event</title>
</head>
<body>
  <header><h1>TechFest 2026</h1></header>
  <p>Registrations are open for all students.</p>
  <p>Participants must carry their identity cards.</p>
  <div>Last date for registration: 15 August 2026.</div>
</body>
</html>
```

Apply the most appropriate type of CSS to achieve the following effects:

- The header style must be reused on every event page: dark-green background, white text, and centered alignment.
- Both paragraphs require 17-pixel font size and justified alignment only on this page.
- Only the notice div requires a light-yellow background and bold red text.

a) Identify whether inline, internal, or external CSS should be used for each effect and justify your choices **(3 marks)**

b) Write the required CSS and show how it should be connected or applied to the HTML **(2 marks)**

Q11. Hospital Information Page

Consider the following HTML code:

```
<!DOCTYPE html>
<html>
<head>
  <title>Hospital Services</title>
</head>
<body>
  <header><h1>CityCare Hospital</h1></header>
```

```
<p>Emergency services are available throughout the day.</p>
<p>Patients may book appointments through the online portal.</p>
<div>Emergency Contact: 108</div>
</body>
</html>
```

Apply the most appropriate type of CSS to achieve the following effects:

- The header design is common to all hospital pages: blue background, white text, and center alignment.
- Both paragraphs need 18-pixel font size and 1.5 line spacing only on this page.
- Only the emergency div needs a red border, light-pink background, and bold text.

a) Identify whether inline, internal, or external CSS should be used for each effect and justify your choices (3 marks)

b) Write the required CSS and show how it should be connected or applied to the HTML (2 marks)

Q12. Online Store Offer Page

Consider the following HTML code:

```
<!DOCTYPE html>
<html>
<head>
  <title>Festival Offers</title>
</head>
<body>
  <header><h1>Festival Mega Sale</h1></header>
  <p>Discounts are available on selected electronics.</p>
  <p>Free delivery is available for orders above ₹1,000.</p>
  <div>Offer ends tonight at 11:59 PM.</div>
</body>
</html>
```

Apply the most appropriate type of CSS to achieve the following effects:

- The same header design will be used across the complete shopping website: black background, white text, and centered alignment.
- Both paragraphs require 16-pixel font size and dark-blue text only on this page.
- Only the offer div requires an orange background, white text, and bold formatting.

a) Identify whether inline, internal, or external CSS should be used for each effect and justify your choices (3 marks)

b) Write the required CSS and show how it should be connected or applied to the HTML (2 marks)

Q13. Marathon Registration

A marathon registration form asks participants to select exactly one race category: 5 km, 10 km, or 21 km. It also allows them to select multiple optional facilities: timing chip, event T-shirt, breakfast, and transport.

a) Write the HTML code for selecting exactly one race category (2 marks)

b) Write the HTML code for selecting multiple optional facilities (2 marks)

c) Explain why radio buttons are suitable for the first requirement and checkboxes for the second (1 marks)

Q14. Student Club Enrollment

A student club portal asks students to select exactly one primary club: Coding Club, Robotics Club, or Design Club. Students may also select multiple activities of interest: workshops, competitions, guest lectures, and project exhibitions.

- a) Write the HTML code for selecting exactly one primary club (2 marks)**
- b) Write the HTML code for selecting multiple activities of interest (2 marks)**
- c) Explain why radio buttons are suitable for the first requirement and checkboxes for the second (1 marks)**

Q15. Conference Participation Form

A conference portal asks delegates to select exactly one participation mode: In-person, Online, or Hybrid. Delegates may choose multiple tracks: Artificial Intelligence, Cybersecurity, Data Engineering, and Web Technologies.

- a) Write the HTML code for selecting exactly one participation mode (2 marks)**
- b) Write the HTML code for selecting multiple conference tracks (2 marks)**
- c) Explain why radio buttons are suitable for the first requirement and checkboxes for the second (1 marks)**